Algorithm 1 Dual-Based WMMSE Algorithm for Calculating $\mathbf{W}$

Input: $\mathbf{c}_k$, $\forall k \in \mathcal{K}$ and μ.

Output: $\mathbf{W}^{\star}$.

1: Initialize $\mu^{(0)}$, $\mathbf{W}^{(0)}$, and iteration index $t = 0$.
2: Calculate the value of the objective function in (19a) as $\mathrm{WSR}\left(\mathbf{W}^{(0)}\right)$
3: **repeat**
4: set $t = t + 1$;
5: Update $g_k^{(t)}$ by (14).
6: Update $\tau_k^{(t)}$ by (17).
7: Update $\mathbf{W}^{(t)}$ by (23) and search $\mu^{(t)}$ by employing bisection method.
8: **until** $\left|\mathrm{WSR}\left(\mathbf{W}^{(t)}\right) - \mathrm{WSR}\left(\mathbf{W}^{(t-1)}\right)\right| \leq \epsilon$.
9: Return $\mathbf{W}^{\star}$.